# WEB-BASED SALES, INVENTORY, CUSTOMER, AND LOCATION MAPPING MANAGEMENT SYSTEM FOR A WATER REFILLING STATION FOR AQUA FABZ WATER REFILLING STATION

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**BACHELOR OF SCIENCE IN INFORMATION SYSTEMS**

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Chapter 1

# INTRODUCTION

## Background of the Study

Water refilling stations have become a common and affordable source of drinking water in many Philippine communities. The growing demand for them has intensified competition among station owners, who now look to effective management systems to run their businesses (Asis et al., n.d.). Most stations are small, owner-managed businesses that handle many small transactions every day, such as refills, containers and supplies, pickup and delivery orders, and vending machine sales. For such a business, accurate records of what is sold, what remains in stock, which containers are with customers, and where orders must be delivered are central to daily decisions. Timely and organized records therefore directly affect a station's service to its customers and its ability to stay competitive.

Aqua Fabz Water Refilling Station is located in Brgy. Enclaro, Crossing Pag-Asa, Binalbagan, Negros Occidental. Besides refilling water, it sells containers, caps, and other supplies, accepts pickup and delivery orders, and operates vending machines. The station records its sales, vending machine sales, stock, containers, orders, payments, and deliveries manually, and customer and delivery information is kept separately. This leads to errors in recording sales and stock, difficulty combining sales from the station and its vending machines, no systematic tracking of borrowed, returned, and damaged containers, late awareness of items that need restocking, and difficulty for delivery personnel in

locating customer addresses. This scattered, manual recording is a bottleneck in tracking daily operations, and without unified records the owner cannot easily see overall sales, stock levels, container movement, or delivery areas.

These conditions mirror other businesses, confirming that they are common weaknesses of manual record-keeping. A local study found that an integrated sales and inventory management system was cheaper, faster, and more efficient than the manual methods used by the company studied (Cambronero et al., 2013), and a web-based sales and inventory system for a business in Davao City sped up the monitoring of sales and inventory records and reduced labor and operating costs (Carado et al., 2013). In Kupang City, Indonesia, drivers delivering tank water did not know exactly where their customers were, which led to a web-based geographic information system for tank water filling locations (Lopes, 2018). For water refilling stations in particular, a management system developed for Far East Water Station in Mabalacat City, Pampanga was well received, with alpha and beta test scores of 3.76 and 4.2 (Asis et al., n.d.). None of these studies, however, combines sales, vending machine sales, inventory, containers, customers, deliveries, and location mapping in one platform for a single station, which is the gap this study addresses.

It is for these reasons that the researchers propose a web-based sales, inventory, customer, and location mapping management system for Aqua Fabz Water Refilling Station. The purpose of this capstone project is to design, develop, and evaluate a centralized platform where staff record sales, orders, payments, customers, containers, and deliveries, where vending machine sales are added automatically to the daily and

overall sales total, where customer delivery locations are shown on an interactive digital map, and where the administrator monitors sales, inventory, and reports for decision-making. The system directly addresses the problems described above by digitizing the workflow the station already follows into one organized process, reducing manual, error-prone, and scattered records. In doing so, it strengthens accountability, shortens the path from order to delivery, and gives the owner reliable records for managing the business.

## Objectives of the Study

### General Objective

The study aims to design and develop the Web-Based Sales, Inventory, Customer, and Location Mapping Management System for a Water Refilling Station for Aqua Fabz Water Refilling Station in Brgy. Enclaro, Crossing Pag-Asa, Binalbagan, Negros Occidental.

### Specific Objectives

**To design the system with the following features:**

a. User registration, login, and staff account management that allows the administrator and staff to securely access the system according to their assigned roles and permissions, and allows the administrator to create, update, and deactivate staff accounts;

b. Customer registration and profile management that allows staff to record and update customer information such as name, contact number, address, customer

type, and multiple delivery locations, and to view each customer's transaction history;

c. Product and product pricing management that allows the administrator to manage products such as purified water, mineral water, containers, caps, bottles, and other station supplies, and to set and update their prices;

d. Sales transaction management and vending machine sales recording that allows staff to record customer purchases, compute totals, and record the sales of each vending machine, which are automatically added to the daily and overall sales total;

e. Order and payment management that allows staff to record pickup and delivery orders with their details and schedules, and to record payment amounts, methods, and status;

f. Container management that tracks available, borrowed, returned, damaged, and customer-owned water containers;

g. Inventory management with stock-in and stock-out recording and low-stock monitoring that updates quantities of containers, caps, bottles, and supplies, including vending machine stock, and alerts users of items that need replenishment;

h. Delivery management and status tracking that records delivery requests, schedules, assigned personnel, and destinations, and tracks orders as Pending, Preparing, Out for Delivery, Delivered, or Cancelled;

i. Location mapping and geolocation that allows staff to record, locate, and visualize customer delivery addresses and the station on an interactive digital map,

helping the administrator monitor delivery areas and giving delivery personnel a visual reference of customer destinations;

j. Dashboard, sales monitoring, and reports generation that displays total sales (including vending machine sales), pending orders, completed deliveries, available containers, low-stock products, and delivery locations, provides daily, weekly, and monthly sales summaries, and generates sales, inventory, customer, product, payment, order, container, delivery, vending machine, and location-based reports; and

k. Search and filtering, printable transaction records, activity logs, and account management that help staff find records quickly, print receipts and records, and allow the administrator to monitor important user actions for accountability and security.

**To evaluate the quality of the system** through an experts' evaluation (5 members).

**To evaluate the acceptability of the system** through an end-users' evaluation (10 members: the administrator, staff, and delivery personnel of Aqua Fabz Water Refilling Station).

**To develop the User Manual** for the system.

## Scope and Limitation

### Scope

The study covers the design, development, and evaluation of a web-based system for managing the sales, inventory, customers, orders, payments, containers, deliveries,

and customer locations of Aqua Fabz Water Refilling Station, including the sales of the vending machines owned by the station. It serves two user levels: the administrator, who manages staff accounts, products, and prices and monitors sales, inventory, and reports, and the staff, who record customers, sales, orders, payments, and deliveries, while delivery personnel use the map references of customer destinations provided by the system. Its outputs are the functional platform with its dashboard, reports, and printable records, the activity logs, and a User Manual. The study is conducted during academic year 2026–2027.

**Limitation**

It is limited to Aqua Fabz Water Refilling Station and its vending machines. It is a web application only, does not include an online payment gateway (payments are only recorded), does not provide real-time GPS tracking of delivery vehicles (the map is used to record and view customer locations), does not read vending machine sales automatically from the machine hardware (sales are entered by staff), and has no customer accounts or online ordering portal. It excludes water purification and water quality monitoring, payroll, and accounting beyond the sales, inventory, and payment records, requires an Internet connection, and is evaluated only by five experts and ten or more end-users, not long-term usage.

## Significance of the Study

The findings of this study contribute new knowledge on how a unified, web-based sales, inventory, customer, and location mapping system addresses the weaknesses of separate, manual record-keeping in a small water refilling business.

The results are useful to the following persons and groups:

**Owner/Administrator.** The owner gains a single place to monitor combined sales from the station and its vending machines, stock levels, containers, and deliveries, and receives reports that support better business decisions.

**Station Staff.** Staff record sales, orders, payments, and customer information faster and with fewer errors, replacing scattered manual records with one organized and searchable system.

**Delivery Personnel.** Delivery personnel get a visual map reference of customer destinations and clear delivery statuses, helping them locate addresses more accurately and reducing time spent searching.

**Customers.** Customers benefit from better organized orders, accurate tracking of their containers and transactions, and more reliable deliveries.

**Carlos Hilado Memorial State University and the College of Computer Studies.** The study adds to the university's body of capstone work and shows the application of systems analysis and design to a real local business.

**Future Researchers.** The system and its documentation serve as a reference for future studies on sales, inventory, and location-based management systems for water refilling stations and similar small businesses.

## Definition of Terms

**Activity Log.** Refers to the system feature that records the important actions performed by the administrator and staff, kept for monitoring, accountability, and security.

**Administrator.** Refers to the owner or authorized user of the system who manages staff accounts, products, and prices, monitors sales and inventory, and uses the reports for decision-making.

**Container Management.** Refers to the system feature that tracks the available, borrowed, returned, damaged, and customer-owned water containers of the station.

**Dashboard.** Refers to the system screen that summarizes total sales (including vending machine sales), pending orders, completed deliveries, available containers, low-stock products, and delivery locations.

**Delivery Management.** Conceptually, delivery management is the process of making deliveries from one location to another effectively and efficiently (Upper, n.d.).

Operationally, it refers to the recording and tracking of delivery requests, schedules, assigned personnel, destinations, and delivery status (Pending, Preparing, Out for Delivery, Delivered, or Cancelled) in the system.

**Delivery Personnel.** Refers to the persons assigned by the station to deliver orders, who use the map references of customer destinations and the delivery statuses provided by the system.

**Geolocation.** Conceptually, geolocation is the determination of the approximate physical location of an object (National Institute of Standards and Technology [NIST], n.d.-a).

Operationally, it refers to identifying a customer's delivery address by its map coordinates in the system.

**Inventory Management.** Conceptually, inventory management is the component of supply chain management that tracks and supervises inventory and stock items (TechTarget, n.d.-a).

Operationally, it refers to the monitoring of the quantity and availability of water containers, caps, bottles, and other station supplies, including stock-in and stock-out recording, container status, and low-stock alerts.

**Location Mapping.** Conceptually, a web map is an interactive display of geographic information that can be opened in a standard web browser (Esri, n.d.).

Operationally, it refers to the display of customer delivery locations and the station on an interactive digital map to help staff and delivery personnel identify destinations and plan deliveries.

**Order Management.** Conceptually, order management is the administration of business processes related to orders for goods or services (TechTarget, n.d.-b).

Operationally, it refers to the recording and following of customer orders for pickup or delivery, including order details, quantities, payments, and requested schedules, until they are completed.

**Role-Based Access.** Conceptually, role-based access control is access control based on user roles, that is, the set of permissions a user receives from the role assigned to them (NIST, n.d.-b).

Operationally, it refers to limiting the system functions available to a user according to the assigned role, either administrator or staff.

**Sales Transaction.** Conceptually, a sales transaction happens when a business sells something to a buyer (Shopify, n.d.).

Operationally, it refers to the record of a customer purchase in the system, including the items bought, quantities, total amount, and date, whether made at the station or through a vending machine.

**Staff.** Refers to the assigned users of the system who record customers, sales, orders, payments, containers, and deliveries.

**User Manual.** Refers to the guide developed by the researchers that explains the features and proper use of the system for its end-users.

**Vending Machine.** Conceptually, a vending machine is a coin-actuated machine through which various goods may be retailed (Encyclopaedia Britannica, n.d.).

Operationally, it refers to the vending machines owned by Aqua Fabz Water Refilling Station, whose sales are recorded per machine in the system and automatically added to the daily and overall sales total.

**Water Refilling Station.** Conceptually, a refilling station is another term for a retail water system establishment, where refilled water is sold and placed in the refilled water container or in the customer's own container (Department of Health [DOH], 1999).

Operationally, it refers to Aqua Fabz Water Refilling Station in Brgy. Enclaro, Crossing Pag-Asa, Binalbagan, Negros Occidental, the business served by the system.

**Web Application.** Refers to software accessed through a web browser rather than an installed program; in this study, the system is developed as a web application that the administrator and staff open in a browser to manage the station's operations.

**Web-Based Sales, Inventory, Customer, and Location Mapping Management System.** Refers to the web-based system designed, developed, and evaluated in this study to manage the sales, inventory, customers, orders, payments, containers, deliveries, and customer locations of Aqua Fabz Water Refilling Station in Brgy. Enclaro, Crossing Pag-Asa, Binalbagan, Negros Occidental.

## Conceptual Framework

The study uses the Input-Process-Output-Outcome (IPOO) framework, a conceptual model commonly used in software development. The input is the data and information given to the system, the process is the set of actions that transforms the input into meaningful results, the output is the product that is generated, and the outcome is the benefit or impact of that product on its target users. Figure 1 shows the framework of the study.

| INPUT | PROCESS | OUTPUT | OUTCOME |
|---|---|---|---|
| • Interview results from the owner and staff of Aqua Fabz Water Refilling Station<br>• Current sales, inventory, customer, order, payment, and delivery processes<br>• Functional and non-functional requirements<br>• Product, price, and customer data<br>• Delivery locations and map data<br>• Hardware and software requirements | • Requirements gathering and analysis<br>• System design (use cases, data flow, database, and interface design)<br>• System development and coding<br>• System testing<br>• Evaluation of quality by experts (5 members)<br>• Evaluation of acceptability by end-users (10+ members)<br>• Revisions and preparation of the User Manual | • Web-Based Sales, Inventory, Customer, and Location Mapping Management System for a Water Refilling Station<br>• Modules: sales, vending machine sales, orders, payments, inventory, containers, customers, deliveries, and location mapping<br>• Dashboard and generated reports<br>• Printable transaction records<br>• User Manual | • Less manual record-keeping and fewer errors in sales and inventory<br>• Consolidated sales from the station and vending machines<br>• Better stock monitoring and timely restocking<br>• More organized customer, container, and delivery management<br>• More efficient delivery planning through map references<br>• Better, data-based decisions by the owner |

↓ *Feedback: revisions based on evaluation results and user suggestions* ↑

*Figure 1. IPOO Framework of the Study*

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